

4.2 FULLY SYNTHETIC: TRAINING ON SYNTHETIC DATA

In this section, we study the fully synthetic setting which corresponds to training solely on synthetic data (i.e. $n = 0$ in equation 6). For simplicity, we consider only label noise and ignore feature noise in the synthetic data. Essentially, this allows us to exhibit the smooth phase transition of the classifier’s accuracy in terms of label noise, which extends the result of Feng et al. (2024). Specifically, we obtain the following corollary of theorem 4.2.

Corollary 4.3 (Performance when training only on synthetic data). *Let $\mathbf{w}_s$ be the Ridge classifier described in equation 6 trained only on synthetic data with only label noise (i.e., $\hat{\mathbf{C}} = \mathbf{I}_p$). Under Assumption 4.1, the decision function $\mathbf{w}_s^\top \mathbf{x}$ on a test sample $\mathbf{x} \in \mathcal{C}_a$ with corresponding label $y = (-1)^a$ and independent of $\mathbf{X}$, satisfies*

$$\mathbf{w}_s^\top \mathbf{x} \xrightarrow{\mathcal{D}} \mathcal{N}(y \cdot \mu_s, \nu_s - \mu_s^2),$$

where

$$\begin{aligned} \mu_s &= \frac{\phi(1 - \varepsilon) - \rho\varepsilon}{\alpha\|\boldsymbol{\mu}\|^2 + \alpha + \gamma(1 + \delta_s)} \|\boldsymbol{\mu}\|^2, \\ \nu_s &= \frac{\lambda^2 \|\boldsymbol{\mu}\|^2}{h(\alpha\|\boldsymbol{\mu}\|^2 + \alpha + \gamma(1 + \delta_s))} \left(\frac{\|\boldsymbol{\mu}\|^2 + 1}{\alpha\|\boldsymbol{\mu}\|^2 + \alpha + \gamma(1 + \delta_s)} - \frac{2(1 - h)}{\alpha} \right) + \frac{1 - h}{h}, \end{aligned}$$

with

$$\eta_s = \lim_{p \rightarrow \infty} \frac{p}{m}, \quad h = 1 - \frac{\alpha\eta_s}{(\alpha + \gamma(1 + \delta))^2}, \quad \delta_s = \frac{\eta_s\alpha - \alpha - \gamma + \sqrt{(\alpha + \gamma - \eta_s\alpha)^2 + 4\eta_s\alpha\gamma}}{2\gamma}.$$

Corollary 4.3 provides an explicit formulation of Theorem 4.2 with synthetic data only and ignoring distribution shift (yielding an explicit expression of δ_s). This setting provides a clearer interpretation of the effect of label noise since the classifier’s performance is directly related to the quantity $\lambda = \phi(1 - \varepsilon) - \rho\varepsilon$. The breaking point of the classifier’s performance occurs at $\lambda = 0$, which corresponds to the accuracy of random guessing, yielding to the critical value of label noise $\varepsilon^* = (1 + \frac{\rho}{\phi})^{-1}$. This critical value is equivalent to the one obtained by Feng et al. (2024), however, we extend their result to the high-dimensional setting which exhibits a smoother phase transition as depicted in Fig. 4. Essentially, the sharp phase transition of Feng et al. (2024) is covered by our result by taking $\eta_s \rightarrow 0$. In this sense, the predicted smooth transition better mirrors real-world scenarios where finite sample sizes introduce gradual changes in performance rather than abrupt shifts. This makes our theoretical findings more applicable and reliable for practical scenarios.

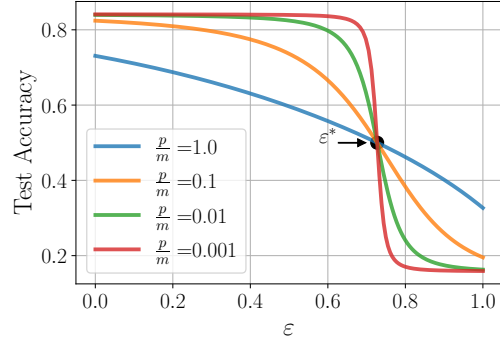


Figure 4: Phase transition in terms of label noise as predicted by Corollary 4.3. The critical value for ε is predicted at $\varepsilon^* = (1 + \frac{\rho}{\phi})^{-1}$. We fix $p = 100$ and vary m . The remaining parameters are $\|\boldsymbol{\mu}\| = 1$, $\rho = 0.3$ and $\phi = 0.8$, i.e. $\varepsilon^* = 0.73$.

5 EXPERIMENTS

In this section, we present our experiments conducted on different real-world tasks and datasets in order to illustrate our theoretical findings presented in the previous section.

5.1 EXPERIMENTAL SETTINGS

Amazon Reviews. We use the Amazon Reviews datasets (Blitzer et al. (2007)) which include several binary classification tasks corresponding to positive versus negative reviews of books,

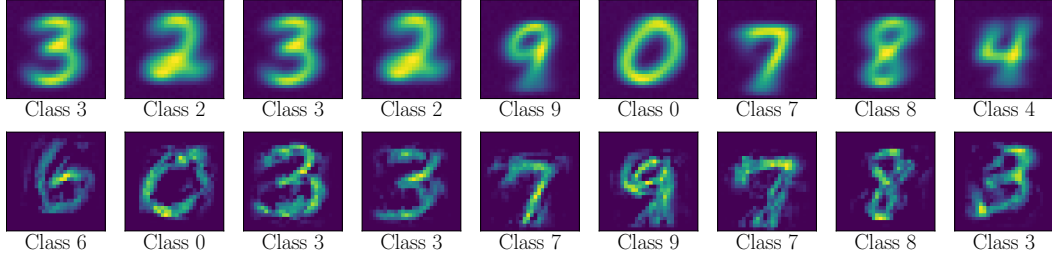


Figure 5: Illustration of two different generation schemes for the MNIST data. *Top figure*: Generating MNIST-like data samples by only estimating the mean of each class $\hat{\mu}_a$ for $a \in [10]$ and without estimating the covariance matrix, i.e samples here are generated through the distribution $\mathcal{N}(\hat{\mu}_a, \mathbf{I}_p)$. *Bottom figure*: Generating samples by estimating both the mean and covariance of each class, as of our considered generative model defined in equation 2.

electronics and kitchen. We apply the standard scaler from `sklearn` (Pedregosa et al., 2011) and estimate $\|\mu\|$ with the normalized data. The synthetic data is generated following the described generative scheme (see equation 2). We use the Ridge classifier in equation 6 for this data.

MNIST. We also conducted experiments on the MNIST (LeCun & Cortes (2010)) dataset to illustrate our theoretical insights, by training a simple neural network with one-hidden layer and ReLU activation function. Concerning the synthetic data, we used different values of $\hat{n}$ to generate new samples in order to highlight the importance of the generation quality, and introduced a label noise ε to emphasize on the importance of the pruning. Figure 5 shows some examples of MNIST-like synthetic data that has been generated and used in our experiments.

LLM Safety Alignment. We also investigated the impact of synthetic text data for the task of alignment of LLMs with direct preference optimization on safety datasets, using the same approach as in (Alami et al., 2024). We finetune the `Falcon 2-11B Instruct` model Malartie et al. (2024) on $n = 5000$ human data from the Anthropic’s *HH-RLHF* dataset², which correspond to real data, while synthetic data are extracted from the PKU safe RLHF dataset³ which are generated using `Alpaca3-70B`⁴. We increase the amount of synthetic data by injecting gradually five batches of 7000 samples per batch, to study the performance of the fine-tuned model as we add more synthetic data. In this experiment, we focus only on label noise by randomly perturbing the synthetic dataset. Each entry from the synthetic dataset includes a prompt $x^{(j)}$, a safe response $y_{sw}^{(j)}$ (safety-accepted response), and a less safe response $y_{st}^{(j)}$ (safety-rejected response). We, therefore, perturbed this dataset by swapping safe and less safe responses with a probability ε (label noise), and selecting the prompts according to a verifier of parameters ρ and ϕ described earlier in this paper.

For the evaluation, we use the `ALERT` dataset⁵ (Tedeschi et al. (2024)) to test the safety of responses of the finetuned model after being judged by `LLama-Guard-3-8B` (Dubey et al., 2024). As in (Alami et al., 2024), we compute the safety score as the percentage of safe answers labeled by `LLama-Guard-3-8B`. We report the results in figure 8 for strong supervision $(\rho, \phi) = (0.2, 0.9)$ and weak supervision $(\rho, \phi) = (0.5, 0.5)$ for both $\varepsilon = 0.1$ and $\varepsilon = 0.5$.

LLM Q&A Safety Generation. This experiment aims to evaluate the impact of synthetically generated prompts (i.e. feature noise). To construct the generative model for this experiment, we fine-tune an LLM (M) with supervised fine-tuning (SFT) on pairs of question-answer (Q&A) sentences extracted from a safety dataset. Initially, we fine-tune M on $12k$ human annotated Q&A as safe or unsafe (Ji et al., 2024), yielding a fine-tuned model on human data denoted as M_h . Then, M_h is considered as the generative model to generate a large dataset of synthetic Q&A prompts

²<https://huggingface.co/datasets/yimingzhang/hh-rlhf-safety>

³<https://huggingface.co/datasets/PKU-Alignment/PKU-SafeRLHF>

⁴<https://huggingface.co/PKU-Alignment/alpaca-70b-reproduced-llama-3>

⁵<https://github.com/Babelscape/ALERT/blob/master/data/alert.jsonl>

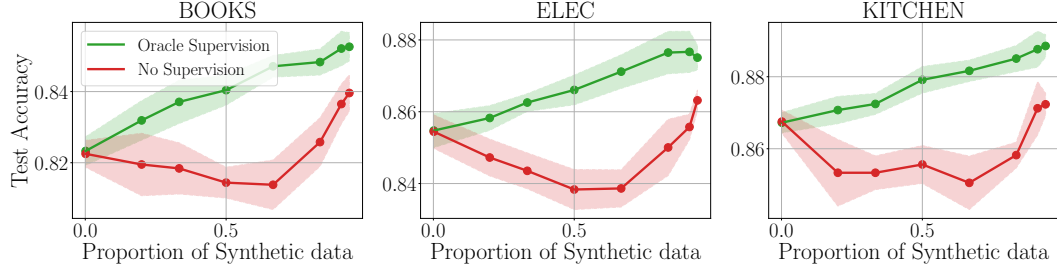


Figure 6: **Results of the Amazon Reviews setting:** Test Accuracy with the proportion of synthetic data evaluated on Amazon Review Blitzer et al. (2007) dataset. The number of real data sample used is $n = 800$, the dimension is $p = 400$, $\gamma = 10^{-1}$ and $\varepsilon = 0.2$ (fixed). The pruning parameters are $(\rho, \phi) = (0, 1)$ for Oracle supervision and $(\rho, \phi) = (1, 1)$ for No supervision.

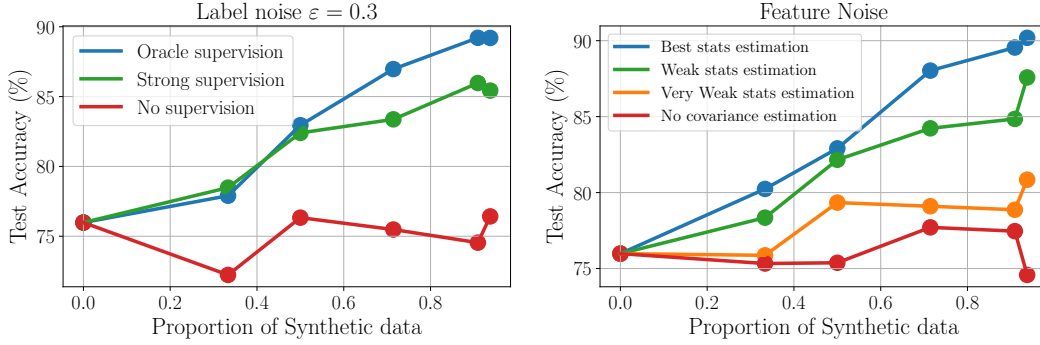


Figure 7: **Results of the MNIST setting:** Training an NN with one hidden layer and ReLU activation function on a mixture of real ($n = 500$) and varying the proportion of synthetic Gaussian data.

(around 120k samples) that were further annotated as safe/unsafe using Mistral-Nemo⁶ and Qwen2-7B-Instruct (Yang et al., 2024), which incorporate a further label noise. To verify the data, we use Llama-Guard-3.1 (Dubey et al., 2024). We conducted this experiment using two LLMs (M) of different sizes (to vary the generative model quality) which are the Llama-3.1-8B and Gemma-2-2B-it (Team et al., 2024) instruct models.

5.2 EFFECT OF LABEL NOISE

Figures 6, 7 (left plot) 8 reflect the effect of label noise. Essentially, as theoretically anticipated, the trained models do not benefit from synthetic data unless it is accurately verified. Specifically, in the case of weak supervision, model performance drops significantly, and the improvement from using synthetic data is only visible with very high synthetic sample sizes. On the contrary, with strong supervision, we observe a monotonous performance boost as the proportion of synthetic data increases.

5.3 EFFECT OF FEATURE NOISE

In this section, we discuss the experiments related to feature noise. In Fig. 7 (right), we depict the performance of a one-hidden layer MLP trained on a mix of real and synthetic MNIST data following our theoretical framework. As we can observe from the figure, the performance boost from synthetic data heavily depends on the generative model quality as predicted by our theoretical results. We further observe the same trend using LLMs as depicted in Fig. 9, where we observe that the synthetic data generated by Llama3.1-8B-Instruct yields a better performance boost compared to Gemma-2-2B-it as we increase the amount of synthetic samples.

⁶<https://mistral.ai/news/mistral-nemo/>